Laboratory Record
of Software Engineering Lab

Roll No.: 160124737112
Exp No.: 01
Sheet No.: _____
Date: _____

SOFTWARE REQUIREMENTS SPECIFICATION

for

ONLINE SHOPPING SYSTEM (OSS)

Version 1.0 Approved

Author:
Vishnu Teja Karampuri
Department of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, Telangana
vishnuteja.karampuri15@gmail.com

July 2026

Laboratory Record
of Software Engineering Lab

Roll No.: 160124737112
Exp No.: 01
Sheet No.: _____
Date: _____

Table of Contents

1. Introduction

(a) Purpose
(b) Document Conventions
(c) Intended Audience and Reading Suggestions
(d) Product Scope
(e) References

2. Overall Description

(a) Product Perspective
(b) Product Functions
(c) User Classes and Characteristics
(d) Operating Environment
(e) Design and Implementation Constraints
(f) User Documentation
(g) Assumptions and Dependencies

3. External Interface Requirements

(a) User Interfaces
(b) Hardware Interfaces
(c) Software Interfaces
(d) Communications Interfaces

4. System Features

(a) User Authentication and Access Control
(b) Product Catalog and Inventory Management
(c) Shopping Cart and Checkout
(d) Order Management and Fulfillment
(e) Reviews and Ratings
(f) Discounts, Coupons, and Promotions
(g) Notifications
(h) Reporting and Analytics

5. Other Nonfunctional Requirements

(a) Performance Requirements
(b) Safety Requirements
(c) Security Requirements
(d) Software Quality Attributes
(e) Business Rules

6. Other Requirements

(a) Data Retention
(b) Legal and Compliance

(c) Internationalization

7. Appendix A: Glossary

8. Appendix B: Analysis Models

9. Appendix C: To Be Determined (TBD) List

Laboratory Record
of Software Engineering Lab

Roll No.: 160124737112
Exp No.: 01
Sheet No.: _____
Date: _____

Revision History

Name	Date	Reason For Changes	Version
Vishnu Teja Karampuri	2026-07-27	Initial Baseline Draft	1.0

Document Index

Key Module / Section	Document Reference
User Authentication & Access Control	Section 4.1
Product Catalog & Inventory	Section 4.2
Shopping Cart & Checkout	Section 4.3
Order Management & Fulfillment	Section 4.4
Reviews and Ratings	Section 4.5
Discounts, Coupons, and Promotions	Section 4.6
Notifications	Section 4.7
Reporting and Analytics	Section 4.8
Performance Requirements	Section 5.1
Security Requirements	Section 5.3

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
స్వయం తేజస్విన్ భవ
1979

Laboratory Record
of Software Engineering Lab

Roll No.: 160124737112
Exp No.: 01
Sheet No.: _____
Date: _____

1 Introduction

1.1 Purpose

This document specifies the software requirements for the development of an Online Shopping System (OSS), a web-based e-commerce platform intended to enable customers to browse products, place orders, and make payments online, while allowing administrators and vendors to manage inventory, orders, and sales operations. This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Version 1.0 of the system and is intended to serve as the primary reference for the design, development, testing, and acceptance of the product.

This document follows the structure recommended by IEEE Std 830-1998, "IEEE Recommended Practice for Software Requirements Specifications." It is written for use by the development team, project stakeholders, quality assurance personnel, and academic evaluators reviewing the project.

1.2 Document Conventions

- Requirement identifiers follow the pattern FR-XX for functional requirements and NFR-XX for non-functional requirements.
- Priority levels are classified as High, Medium, or Low, indicating the criticality of the requirement to the Minimum Viable Product (MVP).
- The keywords shall, should, and may are used per IEEE convention: shall denotes a mandatory requirement, should denotes a recommended requirement, and may denotes an optional requirement.
- Terms defined in the Glossary (Appendix A) are italicized on first use within a section.

1.3 Intended Audience and Reading Suggestions

- **Developers —** should read Chapters 3, 4, and 5 in detail for implementation guidance.
- **Project Evaluators / Reviewers —** should read Chapters 1, 2, and 4 to assess scope and functional coverage.
- **Testers / QA —** should focus on Chapter 4 (System Features) and Chapter 5 (Non-Functional Requirements) to derive test cases.
- **Project Managers —** should read Chapter 2 for scope and Chapter 6 for outstanding items.

Readers unfamiliar with domain-specific terminology should first consult the Glossary in Appendix A.

1.4 Product Scope

The Online Shopping System (OSS) is a centralized, role-based web application designed to manage the complete lifecycle of an online retail operation. The system shall support three primary user roles: Administrator, Vendor/Seller, and Customer, each with a distinct set of permissions and dashboards.

The core objectives of the OSS are to:

- Eliminate manual, in-person or phone-based ordering processes for retail purchases.
- Provide customers with a self-service platform to browse products, manage a shopping cart, and complete purchases securely.
- Provide vendors with tools to list products, manage inventory, and fulfill orders efficiently.
- Provide administrators with a consolidated view of sales, inventory, and customer data for reporting and decision-making.

- Ensure data integrity, transactional consistency, and role-based access control across all modules.

The system does not include point-of-sale (POS) integration for physical retail stores, third-party marketplace syndication, or logistics/fleet management for delivery; these are explicitly out of scope for Version 1.0 and are noted as candidates for future releases.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers, 1998.
- IEEE Std 1016-2009, IEEE Recommended Practice for Software Design Descriptions.
- Node.js Foundation, Node.js Documentation, https://nodejs.org/en/docs.
- MongoDB Inc., MongoDB Manual, https://www.mongodb.com/docs/manual/.
- RFC 7519, JSON Web Token (JWT), Internet Engineering Task Force.
- OWASP Foundation, OWASP Top Ten Web Application Security Risks, https://owasp.org/www-project-top-ten/.
- PCI Security Standards Council, PCI DSS Quick Reference Guide, https://www.pcisecuritystandards.org/.

2 Overall Description

2.1 Product Perspective

The OSS is a new, self-contained software product. It is not a modification of an existing system, although it may later be integrated with third-party services such as payment gateways, shipping/courier APIs, or SMS/email gateways for order notifications.

The system follows a three-tier architecture:

- **Presentation Tier:** A responsive single-page web application (SPA) providing role-specific dashboards (Customer storefront, Vendor console, Admin panel).
- **Application/Logic Tier:** A RESTful API server responsible for business logic, authentication, authorization, cart/order processing, and payment orchestration.
- **Data Tier:** A NoSQL document database responsible for persistent storage of all catalog, order, and account data.

2.2 Product Functions

- User authentication and role-based access control.
- Product catalog and inventory management.
- Shopping cart and wishlist management.
- Order placement, checkout, and payment processing.
- Order tracking and fulfillment management.
- Product reviews and ratings.
- Discount, coupon, and promotion management.
- Notifications (order confirmation, shipping updates).
- Reporting and analytics (sales, inventory, customer behavior).

2.3 User Classes and Characteristics

User Class	Description	Characteristics
Administrator	Manages platform-wide catalog master data, users, vendors, and system configuration.	Technically proficient; requires full access to all modules and reporting tools.
Vendor/Seller	Manages products, stock levels, and fulfills orders for items they list.	Moderate technical proficiency; primarily uses catalog and order-fulfillment modules.
Customer	End consumer purchasing products through the platform.	Varying technical proficiency; requires a simple, intuitive, mobile-friendly interface.

2.4 Operating Environment

- **Client side:** Modern web browsers (Chrome, Firefox, Edge, Safari — latest two major versions), supporting both desktop and mobile viewports.
- **Server side:** Node.js runtime (LTS version) running an Express.js application server, deployable on Linux-based cloud infrastructure.
- **Database:** MongoDB (self-hosted or Atlas-hosted cluster).
- **Network:** HTTPS over the public internet.

2.5 Design and Implementation Constraints

- The system shall be implemented using JavaScript/TypeScript across both frontend and backend to maintain a unified codebase and skill set.
- Authentication shall be implemented using stateless JSON Web Tokens (JWT) rather than server-side sessions, to support horizontal scalability.
- The database schema shall be document-oriented (MongoDB), which constrains the system to eventual-consistency patterns for certain cross-collection aggregate operations (e.g., platform-wide sales reports).
- The system must be deployable on infrastructure with modest resources (minimum 2 vCPU / 4GB RAM) to remain viable for small-to-medium retailers.
- All passwords must be stored using a salted one-way hashing algorithm (bcrypt or equivalent); plaintext password storage is strictly prohibited.
- Payment card data shall never be stored directly by the system; all card transactions shall be delegated to a PCI-DSS compliant third-party payment gateway.

2.6 User Documentation

- A User Manual covering role-specific workflows for Administrators, Vendors, and Customers.
- An API reference document (auto-generated where feasible) for developers integrating with the system.
- An in-application onboarding/help tooltip system for first-time users.

2.7 Assumptions and Dependencies

- It is assumed that end users have access to a stable internet connection and a modern web browser.

- It is assumed that vendors have digitized product catalogs to a degree that permits bulk data import, or accept manual re-entry during onboarding.
- The system depends on a third-party payment gateway for transaction processing; unavailability of this gateway will block checkout but not browsing functionality.
- The system depends on a third-party email/SMS gateway for notification delivery; unavailability of this gateway will degrade, but not block, core functionality.
- It is assumed that a single currency (e.g., INR or USD) is used per deployment instance, simplifying pricing and tax logic.

3 External Interface Requirements

3.1 User Interfaces

- The system shall provide three distinct dashboard layouts corresponding to the Administrator, Vendor, and Customer roles.
- All interfaces shall be responsive and usable on screen widths from 360px (mobile) to 1920px (desktop).
- Form validation errors shall be displayed inline, adjacent to the relevant field, within 200ms of user input where client-side validation applies.
- The system shall provide breadcrumb navigation for any workflow nested more than two levels deep.
- The system shall provide a persistent shopping cart indicator visible from any page in the customer storefront.

3.2 Hardware Interfaces

The system does not directly interface with specialized hardware. Standard peripherals (keyboard, mouse, touchscreen) are assumed for client interaction. No barcode scanners or POS terminals are required for Version 1.0, though such integrations are identified as future enhancements (see Appendix C).

3.3 Software Interfaces

Interface	Purpose	Data Exchanged
MongoDB	Persistent data storage	Product, order, customer, vendor, and payment records (BSON/JSON documents)
Payment Gateway (e.g., Razorpay/Stripe)	Online payment processing	Transaction amount, status, reference ID
Email/SMS Gateway (e.g., SendGrid/Twilio)	Notification delivery	Order/notice text, recipient contact info, delivery status
JWT Library (jsonwebtoken)	Token generation and verification	Signed authentication tokens
Shipping/Courier API (future)	Delivery tracking	Shipment status, tracking ID, estimated delivery date

3.4 Communications Interfaces

- All client-server communication shall occur over HTTPS (TLS 1.2 or higher).

- The API shall expose RESTful endpoints returning JSON-formatted responses with standard HTTP status codes.
- WebSocket or polling-based mechanisms may be used for near-real-time order status notification in future versions.

4 System Features

This chapter enumerates the functional requirements of the OSS, organized by feature area. Each feature includes a description, priority, and a table of associated functional requirements.

4.1 User Authentication and Access Control

4.1.1 Description and Priority

High Priority. Provides secure registration, login, logout, and session management for all user roles, along with role-based authorization for all protected resources.

4.1.2 Stimulus/Response Sequences

- User submits registration or login credentials (email/username and password).
- System verifies credentials, issues a signed JWT, and enforces role-based access on subsequent requests.

4.1.3 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow a Customer to register using an email address and password.	High
FR-02	The system shall allow users to log in using a registered email/username and password.	High
FR-03	The system shall issue a signed JWT access token upon successful authentication, valid for a configurable expiry period.	High
FR-04	The system shall reject access to any protected API endpoint when a valid JWT is not supplied in the request header.	High
FR-05	The system shall enforce role-based access control (RBAC), restricting endpoint access according to the user's assigned role (Admin, Vendor, Customer).	High
FR-06	The system shall allow users to reset a forgotten password via a time-limited email verification link.	Medium
FR-07	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.	Medium

4.2 Product Catalog and Inventory Management

4.2.1 Description and Priority

High Priority. Enables vendors and administrators to manage product listings, categories, and stock levels; enables customers to browse and search the catalog.

4.2.2 Stimulus/Response Sequences

- Vendor creates or updates a product listing with catalog details and stock quantity.
- System validates the listing, assigns a SKU, and makes it available for customer search and purchase.

4.2.3 Functional Requirements

ID	Requirement	Priority
FR-08	The system shall allow a Vendor to create a new product listing with mandatory fields (name, description, category, price, stock quantity, images).	High
FR-09	The system shall automatically generate a unique SKU (Stock Keeping Unit) upon product creation, following a configurable format.	High
FR-10	The system shall allow a Customer to search and filter products by category, price range, and rating.	High
FR-11	The system shall automatically decrement available stock upon successful order placement and prevent orders exceeding available stock.	High
FR-12	The system shall allow a Vendor to update product price, description, and stock levels.	Medium
FR-13	The system shall allow an Administrator to deactivate (soft-delete) a product listing, retaining historical order data.	Medium

4.3 Shopping Cart and Checkout

4.3.1 Description and Priority

High Priority. Allows customers to add products to a cart, apply discounts, and complete a checkout with shipping and payment details.

4.3.2 Stimulus/Response Sequences

- Customer adds, modifies, or removes items in the shopping cart and applies a coupon code.
- System validates stock and coupon eligibility, then processes shipping selection and payment through the gateway.

4.3.3 Functional Requirements

ID	Requirement	Priority
FR-14	The system shall allow a Customer to add one or more products, with a selected quantity, to a persistent shopping cart.	High
FR-15	The system shall allow a Customer to modify item quantities or remove items from the cart prior to checkout.	High
FR-16	The system shall allow a Customer to apply a valid coupon code to the cart, recalculating the order total.	Medium
FR-17	The system shall prevent checkout completion if any cart item's requested quantity exceeds available stock at the time of checkout.	High
FR-18	The system shall allow a Customer to select a shipping address and shipping method during checkout.	High
FR-19	The system shall allow a Customer to complete payment via an integrated payment gateway prior to order confirmation.	High

4.4 Order Management and Fulfillment

4.4.1 Description and Priority

High Priority. Supports order confirmation, tracking, and fulfillment workflows for vendors, customers, and administrators.

4.4.2 Stimulus/Response Sequences

- Customer completes payment and the system generates an order record.

- Vendor updates the fulfillment status as the order progresses through processing, shipping, and delivery.

4.4.3 Functional Requirements

ID	Requirement	Priority
FR-20	The system shall generate a unique order ID and order confirmation upon successful payment.	High
FR-21	The system shall allow a Vendor to update the fulfillment status of an order (e.g., Processing, Shipped, Delivered) for products they supply.	High
FR-22	The system shall allow a Customer to view real-time order status and order history.	High
FR-23	The system shall allow a Customer to request cancellation of an order prior to it being marked as Shipped.	Medium
FR-24	The system shall allow a Customer to initiate a return request within a configurable return window after delivery.	Medium

4.5 Reviews and Ratings

4.5.1 Description and Priority

Medium Priority. Allows customers to submit product reviews and star ratings following a completed purchase.

4.5.2 Stimulus/Response Sequences

- Customer who has purchased a product submits a star rating and written review.
- System recalculates the product's average rating and displays it on the catalog listing.

4.5.3 Functional Requirements

ID	Requirement	Priority
FR-25	The system shall allow a Customer to submit a star rating (1–5) and written review for a product they have purchased.	Medium
FR-26	The system shall calculate and display an average rating for each product based on all submitted reviews.	Medium
FR-27	The system shall allow an Administrator to remove a review that violates content guidelines.	Low

4.6 Discounts, Coupons, and Promotions

4.6.1 Description and Priority

Medium Priority. Allows administrators and vendors to define promotional discounts and coupon codes.

4.6.2 Stimulus/Response Sequences

- Administrator or Vendor defines a coupon code or time-bound discount with validity rules.
- System applies or rejects the coupon/discount automatically at checkout based on those rules.

4.6.3 Functional Requirements

ID	Requirement	Priority
FR-28	The system shall allow an Administrator to define a coupon code with a discount type (percentage or flat amount), validity period, and usage limit.	Medium

FR-29	The system shall prevent the use of an expired or usage-exhausted coupon code at checkout.	Medium
FR-30	The system shall allow a Vendor to configure a time-bound discount on their own product listings.	Low

4.7 Notifications

4.7.1 Description and Priority

Medium Priority. Allows the system to notify customers of order-related events via email/SMS.

4.7.2 Stimulus/Response Sequences

- An order-related event occurs (order confirmation, fulfillment status change).
- System dispatches an email or SMS notification to the Customer, where notifications are enabled.

4.7.3 Functional Requirements

ID	Requirement	Priority
FR-31	The system shall dispatch an email notification to the Customer upon order confirmation.	Medium
FR-32	The system should dispatch a notification when an order's fulfillment status changes (e.g., Shipped, Delivered), if notifications are enabled.	Low

4.8 Reporting and Analytics

4.8.1 Description and Priority

Medium Priority. Provides administrators and vendors with aggregate views and exportable reports across sales and inventory data.

4.8.2 Stimulus/Response Sequences

- Administrator or Vendor requests a sales summary or low-stock inventory report for a specified date range.
- System aggregates the underlying order and inventory data and renders or exports the report.

4.8.3 Functional Requirements

ID	Requirement	Priority
FR-33	The system shall generate a sales summary report for a specified date range, filterable by vendor or category.	Medium
FR-34	The system shall generate a low-stock inventory report highlighting products below a configurable stock threshold.	Medium
FR-35	The system shall allow export of generated reports in CSV format.	Low

5 Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-01	The system shall respond to 95% of API requests within 500ms under a load of up to 500 concurrent users.

NFR-02	The system shall support a catalog of at least 50,000 products without measurable degradation in query response time.
NFR-03	Checkout completion (cart validation through payment confirmation) shall complete within 5 seconds under normal load.

5.2 Safety Requirements

The system does not control physical machinery or safety-critical infrastructure. However, it shall implement safeguards against accidental irreversible data loss: all deletions of catalog and order records (products, orders, customer accounts) shall be implemented as soft deletes, with a recovery window of 30 days before permanent purging.

5.3 Security Requirements

ID	Requirement
NFR-04	All user passwords shall be hashed using bcrypt (or equivalent) with a minimum cost factor of 10 before storage.
NFR-05	All communication between client and server shall be encrypted using TLS 1.2 or higher.
NFR-06	JWT access tokens shall expire within 24 hours; refresh tokens, where implemented, shall be revocable server-side.
NFR-07	The system shall sanitize all user inputs to prevent injection attacks (NoSQL injection, XSS), consistent with OWASP Top Ten guidance.
NFR-08	The system shall never store raw payment card data; all payment processing shall be delegated to a PCI-DSS compliant gateway.
NFR-09	The system shall log all administrative and vendor actions (product deletion, order status override, refund issuance) to an immutable audit trail.

5.4 Software Quality Attributes

- **Availability:** The system shall target 99.5% uptime, excluding scheduled maintenance windows.
- **Maintainability:** The codebase shall follow a modular structure (separated routes, controllers, models) to facilitate independent module updates.
- **Usability:** A first-time user shall be able to complete a core task (e.g., adding a product to the cart) without external instruction, within 3 interactions.
- **Portability:** The backend shall be containerizable (Docker) to support deployment across differing hosting environments.
- **Scalability:** The stateless authentication design shall permit horizontal scaling of the application server tier behind a load balancer, especially during high-traffic sales events.

5.5 Business Rules

- Only an Administrator may create or deactivate vendor accounts.
- A Vendor may only manage products and orders associated with their own catalog.
- A Customer may only view (not modify) order status once an order has been confirmed, except to request cancellation or return.
- Confirmed orders are immutable except via an explicit, logged Administrator or Vendor override (e.g., cancellation, refund).

6 Other Requirements

6.1 Data Retention

Transactional records (orders, payments, invoices) shall be retained for a minimum of 7 years following order completion, in accordance with typical financial record-keeping and tax regulations.

6.2 Legal and Compliance

The system shall be designed to accommodate applicable data protection principles (e.g., limiting collection of personally identifiable information to what is operationally necessary, providing mechanisms for data correction upon request) and shall support PCI-DSS compliant payment handling practices.

6.3 Internationalization

Version 1.0 shall support the English language and a single currency only. The architecture should not preclude the addition of multi-language and multi-currency support in future versions.

7 Appendix A: Glossary

Term	Definition
OSS	Online Shopping System — the software product specified in this document.
JWT	JSON Web Token; a compact, URL-safe means of representing claims for authentication.
RBAC	Role-Based Access Control; an authorization model restricting system access based on user roles.
SKU	Stock Keeping Unit; a unique identifier assigned to a distinct product/variant.
Soft Delete	A deletion strategy in which records are flagged as inactive rather than physically removed from the database.
Cart	A temporary collection of products selected by a customer prior to checkout.
Coupon	A code entitling the bearer to a discount on an order, subject to defined validity rules.

8 Appendix B: Analysis Models

B.1 Use Case Diagram

The primary use cases available to each user role are as follows:

- **Administrator:** Manage Users, Manage Products, Generate Reports.
- **Vendor:** Manage Products, Fulfill Orders.
- **Customer:** Browse Products, Manage Cart, Checkout / Pay, Track Order.

B.2 Data Model Overview

A simplified entity-relationship overview of the core collections in the system: Customer, Product, Vendor, Cart, Order, and Review — where a Customer owns a Cart and places Orders, a Vendor lists

Products, a Cart contains Products, an Order includes Products, and Customers write Reviews about Products.

B.3 Use Case Specification: Checkout and Payment

Field	Description
Use Case ID	UC-04
Name	Checkout and Payment
Actor	Customer
Precondition	Customer is logged in and has at least one item in their cart.
Main Flow	1. Customer reviews cart contents and proceeds to checkout. 2. System displays shipping address and shipping method options. 3. Customer selects/enters shipping details. 4. Customer proceeds to payment and selects a payment method. 5. System validates stock availability for all cart items, then redirects to the payment gateway. 6. Upon successful payment confirmation, the system creates the order record and clears the cart.
Alternate Flow	If any cart item is out of stock at validation time, the system displays an error and prompts the Customer to adjust the cart before proceeding.
Postcondition	Order record is persisted with status “Confirmed”; stock levels are decremented; order confirmation notification is dispatched.

9 Appendix C: To Be Determined (TBD) List

TBD ID	Description	Status
TBD-01	Final decision on third-party payment gateway provider(s) to be integrated.	Pending Evaluation
TBD-02	Whether multi-vendor marketplace commission/settlement logic will be included in Version 1.0 or deferred.	Under Review
TBD-03	Exact data retention and deletion policy pending institutional/legal review.	Under Review
TBD-04	Whether multi-currency and multi-language support will be prioritized for Version 2.0.	Pending Evaluation
TBD-05	Whether real-time shipment tracking via courier API integration will be included in a future release.	Under Review

INSTITUTE OF TECHNOLOGY
స్వయం తేజస్విన్ భవ
1979